UiPath™
AGENTIC AUTOMATION

UiPath Maestro
BPMN 2.0 constructs

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Introduction to BPMN 2.0 constructs

In BPMN 2.0, standard constructs can be grouped into several core categories to represent different aspects of processes.

Flow objects

Events, activities, and gateways, the core building blocks of BPMN diagrams.

Flows

These connect flow objects and determine sequence or flows.

Data

Data refers to the information used, created, or stored during a process.

Markers

Graphical modifiers to enrich the meaning of flow objects and specify behavior.

Task types

Shows who or what performs the work in a process, whether it's a person, agent, system, or robot.

How to use this deck?

This deck is your quick-reference guide to the core building blocks of process modeling with BPMN 2.0.

Each slide gives you a simple explanation, BPMN icon, and examples to help you recognize and apply BPMN constructs, whether you're designing a process, reviewing one, or explaining it to others.

Use it as:

- A learning tool for yourself or your team
- A cheat sheet during process modeling
- A shared resource with customers, partners or UiPath community

Events
Things that *happen* during the process

	Start			Intermediate				End
	Standard	Subprocess interrupting	Subprocess non-interrupting	Catching	Boundary interrupting	Boundary non-interrupting	Throwing	Standard
Plain: Generic event with no visual indicators; used for start, end, or milestones.								
Message: Represents external communication								
Timer: Indicates a specific time or duration delay.								

- Use an **event** to show something that happens in the process.
- Start with a **start event**, like when a customer submits an online order.
- Add an **intermediate event** if something needs to wait or respond, like pausing for 3 days before sending a reminder email.
- End with an **end event**, like sending an order confirmation to the customer.
- Events help you mark the beginning, waiting points, and end of a process.

Activities
Work that is *performed* in the process

Task: An action performed by a person or software within a process.	Task
Subprocess: A group of related tasks bundled together into a single activity to make the main process easier to read.	Subprocess
Call activity: A wrapper that invokes another BPMN process (or global task) stored outside the current diagram.	Call activity

- Use activities when something (whether it's an AI agent, robot, or person) performs work.
- For example, packing items into a box is considered a **task**.
- Or when an agent handles the entire payment process, which includes multiple steps like validating the card and processing the payment. This is called a **sub-process**.
- Use **subprocesses** when your model starts to feel crowded.

Task types
Keep processes *clear and executable*

- Use task types to clearly show what kind of work is being done. This helps everyone quickly understand who or what is responsible.

- For example, use a **User Task** when a person needs to do something (like review an application).
- Use a **Service Task** when a system performs an automated action (like running an agent or a robot).
- Use a **Script Task** when the system runs a script without human input.

Send task: Sends a message to an external entity.	Send task
Receive task: Waits for a message from an external entity.	Receive task
User task: Performed by a person with software interaction.	User task
Manual task: Performed by a person without software.	Manual task
Business rule task: Executes logic via a business rules engine (DMN).	Business rule task
Service task: Fully automated activity.	Service task
Script task: Runs a script automatically when triggered.	Script task

Gateways

Decision points that *govern and control* the process flow

Exclusive gateway (XOR): The classic ‘fork in the road’. Choose one path based on a decision.	
Parallel gateway (AND): Do all paths at the same time.	
Inclusive gateway (OR): Do one or more paths, depending on conditions.	
Event-based gateway: Wait for one event to happen, then follow the path linked to that event.	
Complex gateway: When the standard gateways above can’t be used, you can use custom logic to decide what happens next.	

- You can use a gateway when the process needs to make a decision or split into multiple paths.
- For example, you might check whether payment was successful to decide what happens next. This is an **exclusive gateway**.
- You might also need to do two things at the same time, like preparing an invoice and updating inventory. This is a **parallel gateway**.
- Or you might check which delivery options the customer selected and follow one or more paths based on their choices. This would be an **inclusive gateway**.

Markers
Used to specify *behavior* of objects

- You can use markers when you want to show how an activity behaves.
- For example, if a robot needs to send the same email to 100 customers, you can use a **multi-instance sequential marker** to show that the task repeats for each customer.
- Markers help clarify *how* the work happens, not just *what* the work is.

Multi instance parallel marker: Runs the task multiple times in parallel.	
Multi-instance sequential marker: Runs the task multiple times in sequence.	

Flows
Keep processes *connected*

Sequence flows: Shows the order of steps in a process.	
Message flows: Shows communication between different people, systems, or roles.	
Association: Links extra information, like data or notes, to a task or event.	

- Use flows to show how different parts of your process connect and move forward.
- For example, if a customer places an order, and then the system checks inventory, you would use a **sequence flow** to connect those two steps.
- If the system sends a confirmation message to the customer, you would use a **message flow** to show that communication between the system and the customer.
- And if you want to link a note or a document to a task, like attaching shipping instructions to “Pack the order,” you’d use an **association**.

Data

Shows the *information used or produced* during a process.

- In a hiring process, when a candidate applies, the system creates a “Job Application” **data object**. This application is then reviewed in the next step.
- All past job applications are saved in a “Candidate Database” **data store**. This lets the HR team search and reuse data across different hiring processes.

Data object: A piece of information used or created during a process	
Data store: A place where data is saved and can be accessed across processes	

How to get started:

Go to Studio Web:

- **Model from scratch:** Create a new process manually using the modeling canvas.
- **Import a BPMN file:** Upload an existing BPMN model to continue working on it.
- **Use Autopilot:** Describe your process in plain language or upload a diagram image—Autopilot will help generate the model for you.

Click here for more documentation

THANK YOU.

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